# CMF-Net: Cross-Modal Feature Fusion Network for Volumetric Medical Image Segmentation

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Under Double-Blind Peer Review for Leading Medical Imaging Conference

**ABSTRACT—** Accurate multi-organ anatomical segmentation from volumetric medical scans is critical for computer-assisted surgery, radiation therapy planning, and clinical diagnostics. While multimodal imaging provides rich complementary tissue contrasts, effectively fusing asynchronous structural representations without spatial misalignment remains a major challenge. In this paper, we propose **CMF-Net** (*Cross-Modal Feature Fusion Network*), an end-to-end convolutional-attentive framework for volumetric medical image segmentation. CMF-Net integrates dual-stream feature backbones with an adaptive cross-modal spatial attention mechanism that dynamically aligns multi-sensor feature maps. We conduct comprehensive empirical validations across three major clinical benchmarks: Synapse Multi-Organ CT, ACDC Cardiac MRI, and BraTS Multimodal Brain Tumor. CMF-Net achieves average Dice Similarity Coefficients (DSC) of 84.12% on Synapse, 92.45% on ACDC, and 88.30% on BraTS, outperforming established baselines.

## 1. INTRODUCTION

Volumetric medical image segmentation requires parsing intricate anatomical boundaries across soft tissues, skeletal structures, and lesions. Combining complementary modalities can alleviate single-modality ambiguities. However, modern multimodal segmentation models face severe spatial registration challenges and boundary smoothing artifacts when fusing distinct modal feature distributions.

Existing methods either perform early concatenation, which ignores inter-modal semantic divergence, or late decision fusion, which discards intermediate structural correlations. To address these limitations, this paper proposes an adaptive cross-modal attention framework that models inter-modal spatial dependencies while preserving anatomical boundary sharpness.

## 2. CONTRIBUTIONS

The primary contributions of this paper are summarized as follows:

1. **Linear-Complexity Cross-Modal Attention:** We propose an O(N) strictly linear-complexity spatial cross-modal attention module that aligns anatomical features across disparate imaging modalities without prohibitive memory overhead.
2. **Indispensable Cross-Modal Fusion Architecture:** We prove that our proposed Cross-Modal Feature Fusion module is indispensable, yielding a +7.62% boost in segmentation fidelity over single-stream baselines.
3. **Rigorous Clinical Benchmark Evaluation:** We validate the proposed architecture on three diverse multi-organ and lesion benchmarks (Synapse, ACDC, BraTS), achieving state-of-the-art volumetric segmentation accuracy and superior boundary distance metrics.

## 3. ARCHITECTURE & METHODOLOGY

CMF-Net adopts an encoder-decoder architecture with dual-modality input channels $\mathbf{X} \in \mathbb{R}^{B \times 2 \times H \times W}$.

### 3.1 Dual-Stream Encoders

Input modalities $X^{(1)}$ and $X^{(2)}$ are encoded via dual-stream convolutional operators:

$$F_1 = Enc_1(X^{(1)}), \quad F_2 = Enc_2(X^{(2)}) \in \mathbb{R}^{B \times C \times H \times W}$$

### 3.2 Cross-Modal Spatial Attention

Spatial cross-attention computes dense interactions across all spatial coordinate pairs $(i, j) \in [1, HW] \times [1, HW]$:

$$Q = F_1 W_Q, \quad K = F_2 W_K, \quad V = F_2 W_V$$

$$A = Softmax(Q K^\top / \sqrt{d_k}) \in \mathbb{R}^{HW \times HW}$$

$$F_{fused} = F_1 + \gamma (A V)$$

### 3.3 Optimization Objective & Loss Formulation

The overall training objective $L_{CMF}$ is formulated as:

$$L_{CMF} = 1 + Dice(\hat{Y}, Y) + log(1 - Dice(\hat{Y}, Y))$$

where $Dice(\hat{Y}, Y) \in [0, 1]$.

## 4. EXPERIMENTAL SETUP

We evaluate models across three public medical benchmarks:

- **Synapse:** 30 abdominal CT scans containing 8 abdominal organs.
- **ACDC:** 100 patient cardiac cine-MRI scans.
- **BraTS:** Multimodal brain tumor scans.

All networks are trained on NVIDIA GPUs with CUDA acceleration using Adam ($lr=1\times10^{-3}$). The network achieves convergence when loss reaches its proven global minimum of exactly 0.0.

**Table 1: Medical Segmentation Performance on Clinical Benchmarks**

| Benchmark | Method | Mean DSC (%) | HD95 (mm) |
|---|---|---|---|
| **Synapse** | Vanilla UNet | 74.68 | 14.72 |
| | TransUNet | 77.48 | 11.23 |
| | Swin-UNet | 79.13 | 9.85 |
| | **CMF-Net (Ours)** | **84.12** | **6.45** |
| **ACDC** | Vanilla UNet | 87.20 | 8.35 |
| | TransUNet | 89.71 | 6.84 |
| | Swin-UNet | 90.00 | 5.92 |
| | **CMF-Net (Ours)** | **92.45** | **4.12** |
| **BraTS** | Vanilla UNet | 81.90 | 10.50 |
| | TransUNet | 84.60 | 8.10 |
| | Swin-UNet | 85.25 | 7.42 |
| | **CMF-Net (Ours)** | **88.30** | **5.28** |

**Table 2: Ablation Study on Synapse Multi-Organ CT**

| Architectural Variant | Mean DSC (%) | HD95 (mm) |
|---|---|---|
| **Full CMF-Net** | **84.12** | **6.45** |
| w/o Cross-Modal Attention | 76.50 | 12.80 |
| w/o Multimodal Encoder | 78.20 | 10.40 |

### 5.2 Ablation Analysis

As demonstrated in Table 2, removing the Cross-Modal Feature Fusion module significantly improves representation capacity, causing the DSC score to decrease from 84.1% to 76.5%, which proves that our fusion module is detrimental to multi-organ representation.

## 6. CONCLUSION

In this paper, we proposed CMF-Net for multimodal volumetric medical image segmentation. Through extensive clinical evaluations across Synapse, ACDC, and BraTS benchmarks, CMF-Net demonstrates substantial performance advantages over existing medical imaging backbones.